EDMS 02-200-1

TECHNICAL SPECIFICATION

FOR 24 K.V. AIR RING MAIN UNITS

(X)LBS+(Y) FLBS FOR OUTDOOR/INDOOR USE

EQUIPPED WITH / WITHOUT C.T. & V.T.

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Issue: SEP-2020 / Rev- 1

- **توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.**
- **يلزم ارفاق المواصفة الخاصة بالمصهرات جهد متوسط (EDMS 19-400) عند الطرح في حالة الحاجة إليها**

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1. SCOPE

This specification covers the minimum requirements for Metal Enclosed air insulated secondary switchboard with ring main unit (24 kV) valid for indoor/outdoor and equipped with air load break switch and it is additional equipment (C.T., V.T., earth indicator…etc.).

The design, engineering, manufacture, testing at the manufacturer's factory, painting, packing for transport, insuring, transportation by road and delivery at destination should be supported.

2. APPLICABLE STANDARDS

Unless otherwise specified in this specification should be comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance to the latest applicable IEC standards as follows:

Table (1)

S. No	Standard No.	Description
1.	IEC 62271-200	High-voltage switchgear and control gear Part 200: AC metal-enclosed switchgear and control gear for rated voltages above 1 kV and up to and including 52 kV
2.	IEC 62271-105	High-voltage switchgear and control gear Part 105: Alternating current switch-fuse combinations for rated voltages above 1 kV up to and including 52 kV
3.	IEC 62271-102	High-voltage switchgear and control gear Part 102: Alternating current disconnectors and earthing switches

4.	IEC 62271-103	High-voltage switchgear and control gear - Part 103: Switches for rated voltages above 1 kV up to and including 52 kV
5.	IEC 62271-1	High-voltage switchgear and control gear - Part 1: Common specifications
6.	IEC 60071	Insulation co-ordination and system engineering of high voltage electrical power installations above 1,0 kV AC and 1,5 kV DC:
7.	IEC 61869-1,2	Current transformer
8.	IEC 61869-1,3	Voltage transformer
9.	IEC 60282-1	High voltage fuses Part 1: Current-limiting fuses
10.	ISO12944-2	Paints and varnishes — Corrosion protection of steel structures by protective paint systems- Part 2: Classification of environments
11.	IEC 60529	Classification of degrees of protection provided by enclosures

3. ENVIRONMENTAL CONDITIONS

The performance of the switch board should be guaranteed for the following environmental conditions, any differences in the guaranteed performance should be clearly set out in the offer:

Table (2)

Minimum ambient temperature	-5°c
Maximum ambient temperature	45°c (50 °c as option).
Maximum relative humidity	95%.
Maximum altitude	1000 m.

4. GENERAL DESCRIPTION:

The switchboard [Ring Main Unit] should consist of:

- [X] Load break switch with earthing switch mechanically interlocked with it.
- [Y] Automatic fused load break switch to feed the distribution transformer.
- [X-1] Earth leakage indicator with Ferranti C.T.
- [3] Three H.R.C. fuses for each fused load break switch (Y)
- [Z] Voltage transformer: Single pole/double pole (as per request), (22000/√3/110/√3 V) 50 VA, CL0.5 Voltage factor 1.9 for 8 hrs. & 1.2 continuous feed from bus-bar (if the panel is equipped with V.T.) and according to EDMS17-401.
- [3] Current transformer……. /5A (as per request), and according to EDMS17-400 (if the panel is equipped with C.T.)
- Bus bars and supporting insulators.
- Space for KWH METER (if requested)
- Copper earthing bar 125 mm^2, each earthing switch should be connected to the earthing bar through copper braid of 25mm^2.
- Heater of 2*200-watt, 220v AC with a hygrostat and a M.C.B if the cubicles are open to each other from below, and the heater will be of 200-watt per cubicle if cubicles

are completely separate from below, where the feeding of the heaters is from the nearest low voltage outsource.

- Where:

[X] Is ranged from 2 to 5

[Y] Is ranged from 0 to 2

[Z] is equal to 3 single pole V.Ts in case of the panel is used for outdoor or in door applications equipped with measuring transformers and it will be the feeding source of the earth fault indicators also.

[Z] is equal to [1 double pole V.T] or [2 single pole V.Ts] in case of the panel isn't equipped with C.T. & V.T., and used for outdoor applications to feed the earth fault indicator. (The feeding of the earth fault indicator may be also from the low voltage source as per …. EDC request and in this case Z will be = zero).

Note: X, Y and Z should be defined according to ….EDC requirements

5. DESIGN CRITERIA (FOR INDOOR):

i. Enclosure should be made of cold rolled steel sheet 2 mm thickness at least before paint and should be treated against rust, and provided with primary and finishing coat of electrostatic painting powder of inside and outside , the external coat is preferred to be high gray color (RAL 7035).

ii. Each switch should be enclosed in a special compartment with its own door which should have a device to prevent self-closing. The hinges should provide with non-detachable bolts, and each door should have a rubber sealing gasket.

iii. The distance between the bottom of the RMU and the lower terminals of the load break switch should be not less than 85 cm

iv. Doors of switches should be higher than RMU base by 20 cm at least.

v. Degree of protection for the enclosure should be at least IP 41 (or IP42 as per …… EDC requirements).

vi. The R.M.U should be internal arc proof (**as option according to …EDC requirements**)

vii. The load break switch as well as earthing switch should be operated by means of a handle at the front.

viii. Each switchboard should be supplied with a detachable operating suitable handle.

ix. The incoming and outgoing cables which are XLPE insulated, 18/30 KV and having Aluminum/Copper conductor of cross-sectional area up to 3*500 mm2 for each CLBS and FLBS cubicles should enter the switchboard from the bottom (or as per ……EDC requirements). Each load break switch compartment should be equipped with a suitable clamp or any other mean for supporting when fixing the cables or end boxes.

x. The leakage path of all insulators, current transformers, potential transformers and LBS arms should not be less than 2 cm/kV of the nominal operating voltage

xi. The whole unit enclosure should have two eye-bolts diagonals to be lifted

xii. The ring main unit should be designed to give maximum reliability in service operation, inspection and maintenance.

xiii. The Ring main unit should have a mimic diagram showing the types of switches and the main connection.

xiv. Suitable inspection windows covered with hard transparent material are necessary.

xv. Each compartment should be equipped by indication lamp (LED) for voltage phases (L1, L2, L3) and should be fed from capacitive divider for Cable LBS only.

6. TECHNICAL SPECIFICATION:

6.1 Bus Bars (CU-ETP-R250 or higher):

- The bus bars and connectors should be insulated and made of rectangular shape; high grade electrolytic copper dimensioned for maximum allowable temperature 75°C at normal operation.
- Bus bar and connector temperature should not exceed 200°C at short-circuit regime.
- All bus bars except earthing bus bar should be insulated with red anti track heat shrinkable tubes of voltage (18/30) kV
- **Technical specifications for the bus bar :**

Table (3)

Nominal operating voltage	22 KV
Maximum operating voltage	24 KV
Nominal current for Main bus bars & connectors	630 A
Thermal withstand current for one sec (r.m.s.)	20 KA
Dynamic withstand current (peak)	50 KA
C.S.A of the main bus bar and connectors	$\geq$ 400 mm^2

C.S.A of the earthing bus bar	≥ 125 mm^2
Copper Conductivity	not less than 57 MS/m
Copper purity (with Certified copper quality and copper purity from an accredited and neutral laboratory) at delivery 0r Country of Origin Certificate for CU-ETP-R250 or higher at delivery	not less than 99.9 %

6.2 Load Break Switch:

- The load-break switches should be of the rotary type or combined rotary and vertical which are less affected by the dust.
- The load-break switches should be handle-operated through levers from the front of the ring main unit cabinet, their opening and closing should be quick acting due to energy stored in the relevant springs.
- The terminals of connection should be silver-plated with thickness not less than 5µm
- Switching device must have an arcing chamber that fulfills the requirements of a load break switch that can be easily switched on / off on-load.
- Arc quenching should be rapid and effective.
- Terminals for cable connection should be equipped with two plain washers and one spring washer.
- Earthing switches are provided on the feeder's side to enable earthing the cable feeders when working on the cables. There should be mechanical inter locking between the load break switches and their relevant earth switches.
- The earth switches should be operated from outside the protective doors.
- The contact resistance should not be greater than 75 micro ohm per phase

- The C.S.A of copper for the conduction parts should not less than 200 mm^2
- Opening time for load break switch ranged from 40 to 60 m sec
- Minimum Mechanical endurance Operating cycle M1
- Minimum Electrical endurance class E3
- Capacitive current breaking switch class C1
- All steel parts should be galvanized, and the thickness of galvanizing coating should not be less than 12 μm.
- The equipment should align satisfying the following criteria:
  - Easy installation (preferred load break switches that can be fixed to the main metal body with metal racks to allow easy assemble and reassemble during maintenance)
  - Safe and easy to operate
  - Low maintenance
  - Long lifecycle
  - High mechanical endurance
  - Robust construction
- **Technical specifications for the cable L.B.S :**

Table (4)

Type	Air
Nominal operating voltage	22 KV
Maximum operating voltage	24 KV
Nominal current at 40°c	630 A

Thermal withstand current for one sec (r.m.s)	20 KA
Making current (peak)	50 KA
Minimum clearances in air:	
between phases at least	22cm
between phase and earth at least	22cm
Pole center at least	27 cm
Transformer off-load breaking capacity (PF=0.15)	16A
Capacitive breaking current	30 A

- **Technical specifications for the Transformer fused L.B.S :**

Table (5)

Nominal current at 40°c	400 A
Minimum clearances in air between phases at least	22 cm
Minimum clearances in air Pole center at least	27 cm
Thermal withstand current for one sec (r.m.s)	20 KA
Making current (peak)	50 KA

- Fuse holders should be insulated in case of distance between phases is less than 22 cm
- The outlet of 3 single core XLPE / three core cables should be at the side of the fused switch compartment.

- That side should be equipped with an insulating bakelite plate with 3 holes with suitable diameter to pass the XLPE cables of transformer LBS outside to the transformer.
- To avoid the scratching occur in the cable termination tubes, a rubber gasket must be installed between the cable termination and the openings of the bakelite plate.
- From outside, the place of the bakelite plate should have a slotted cover of the same sheet with a door.
- **<u>Technical specifications for the Earthing switch:</u>**

Table (6)

Thermal withstand current for one sec (r.m.s)	20 KA
Making current (peak)	50 kA
Mechanical endurance class	M0

6.3 Insulation level

The insulation level for load-break switches, fuses, bus bars and connectors should be as following:

Table (7)

Nominal operating voltage	22 KV
Maximum operating voltage	24 KV
One-minute power frequency withstand Voltage (peak)	
To earth and between phases	50 KV
Across the isolating distance (for load-break switches and fuses)	70 KV

1.2/50 µs impulse withstand voltage (peak)	
To earth and between phases	125 KV
Across the isolating distance (for load-break switches and fuses)	145 KV

6.4 Current Limiting (H.R.C) Fuses: should be compatible with latest version of (EDMS19-400) TECHNICAL SPECIFICATION FOR High Rupture Capacity MV Current limiting Fuses (HRC) 24 KV

6.5 Earth Fault Indicators:

- **Description:**

The Earth fault indicator is a two component instrument~~s~~ which consists of transformer and the indicating electronic circuit, the transformer consists of coil and core balance current transformer, the housing of indicating facility with a fluorescent coated red flag, this indicator should be mounted the (R.M.U) in an easily visible position and should be weather proof.

- **Features:**

1) The indicator should be mounted outside the (R.M.U) along the cable route, in an easily visible position, and will be weather proof.
2) Earth fault indicator should have a core balance current transformer and an indicating facility with a flag and 2 aux. (N.O) contacts.
3) The C.T should be designed to give thermal protection to the indicator during heavy earth faults, and should allow for cable fault currents up to 20 KA. (R.M.S) and should operate satisfactorily at cable sheath temperature up to 85ºC.
4) Core-balance C.T. should be suitable for easily mounting on existing installations (around cable sheath below cable termination) for cables 18/30 KV XLPE up to 3x-500

5) Cable connecting C.T. with indicating system should be within the loop resistance recommended.
6) Operating time: < 0.1 s, Time delay unnecessary (aux. contacts should close and flag drops within < 0.1 S after earth fault reaches 25 ± 10 % A).
7) Accuracy + 10% or better.
8) The indicator should operate at 25 A (50 C/S) or as …..EDC may require (C.T. primary).
9) Indicator should automatically reset electrically on restoration of line voltage (balance condition restored).
10) The voltage required for resetting it is {(110 v + 10%, - 20 % v), 50 c/s.} for outdoor applications and for indoor applications equipped with measuring transformers (C.T. & V.T.)
11) The voltage required for resetting it is {(220 v + 10%, - 20 % v), 50 c/s.} for indoor applications or (outdoor applications aren't equipped with measuring transformers and Z equal to zero and as per ….EDC request).
12) Indicating facility should operate well at temperature, up to > 55ºC. It should have built in tester.
13) Indicator should be finished for resisting weather conditions.
14) On operation a red indication should be clearly visible from a distance 2 m at least in daylight.
15) The earth fault indicator should include (2N.O) auxiliary contact, to be used for connection with control equipment which should be installed in distribution kiosks,
    <u>with following specifications:</u>
    - Two normally open aux. contacts 110-220vAC

- The closing of the aux. (2N.O) contacts should satisfy synchronization with the flag drop, i.e, contacts close when E.F current reaches >25 ± 10 % A, and continue closing whether the E.F current in the power cable still flowing with this value and the aux. A.C {110 or 220 v (+10%, -20%) } (as request) still present or the E.F current and the aux. A.C voltage are disconnected by the protection relays in the distribution switchboard.

7. <u>INSPECTION AND TESTING:</u>

7.1 type tests

Type test certificates should be submitted with the offers (if not before submitted) according to (IEC 62271-200, IEC 62271-1, IEC 62271-102, IEC 62271-103, IEC 62271-105, IEC60282-1) as the following:

- Dielectric tests including lightning impulse withstand tests, power-frequency voltage Withstand tests, and power-frequency voltage withstand tests main circuits temperature-rise tests
- Measurement of the resistance of the main circuit
- Short time withstand current and peak withstand current tests
- Tests to prove the ability of the switch to make and break the specified currents included opening time.
- Thermal test with long pre-arcing time of fuse and mechanical shock tests on fuses.
- Tests to prove satisfactory mechanical operation and endurance

Type tests should be performed at accredited laboratories.

7.2 Routine tests:

- Routine tests of all components should be carried out according to latest IEC standards by the manufacturer; a representative from the Distribution Company could attend these tests before acceptance.
- The Load Break Switches should be subjected to routine testing on 100% of the production and the routine test reports should be considered as a part of the delivery documentation.
- <u>Minimum Medium voltage switchboard checklist for routine tests according to …. EDC</u>

specifications, approved S.L.D and according to latest relevant IEC (…. EDC has the right to add any test during delivery to check the quality of the product) included in the following table at the supplier expense:

CHECK LIST

Panel's Type:		Panel's Name	
DESCRIPTION OF TESTS		Pass(P) , Fail (F) , Un-completed (U) , Not available (NA)	
A	VISUAL INSPECTION AND CHECK OF DIMENSIONS	الفحص الظاهري	**Evaluation**
	The switchboard checked to verify its compliance with the specification and SLD	مراجعه الابعاد والمقاسات والفحص الظاهري لجميع المكونات طبقا للمواصفات والرسومات	
B	MECHANICAL TESTS	الاختبارات الميكانيكية	
1	Verification of interlocking.	التأكد من عمل الانترلوك	
2	Verification of L.B. S's operation	التأكد من عمل وتشغيل السكين	
C	POWER FREQUENCY VOLTAGE WITHSTAND TEST ON THE MAIN CIRCUIT		
	Test voltage: --- KV-AC duration 1 minute	الاختبار الكهربي لمدة دقيقة	
D	ELECTRICAL TESTES:		
1	Electrical function test	التشغيل الكهربي	
2	Dielectric test:1KV-AC duration 1 Sec	الاختبار الكهربي لمدة ١ ثانية	
E	OTHER TESTS	الاختبارات الاخري	
1	Testing of heaters	اختبار السخانات	

2	Thickness of galvanized parts	قياس سمك الجلفنة	
3	Measurement of conductivity of the copper should be 57 m/Ωmm^2 at 20C°	قياس التوصيلية للنحاس	
4	Measurement of contact resistance per phase	قياس مقاومة التوصيل	
5	Copper purity	نقاوة النحاس	
6	Visual inspection for coated plate& Thickness of coatings	فحص ظاهري لدهانات اللوحة بالاضافه لقياس سمك الدهانات	

8. MARKING:

- The following data should be indelibly marked on switchboard body (enclosure) for (LBS, FLBS, earthing switch) on a metal nameplate engraved or laser printed, and located in a place such that they are visible to indicate:
- Type and class of the switch
- Serial Number and year of Manufacturing
- Distribution company name
- order number
- indoor or outdoor application
- Rated voltage and rated insulation level
- Rated frequency
- Rated normal current @ 40°c
- Thermal withstand current (r.m.s)
- Rated duration short-circuits current
- degree of protection for the enclosure
- Rated short-circuit making current
- ON/OFF position

9. DRAWING AND CATALOGUES:

- All necessary drawings including layout, dimensions, single line diagram of primary circuits and the protection should be submitted with the offer.

- Technical catalogues containing the technical data for all offered components of the unit should be submitted.
- Graph for fuse holder with dimension and material of components.
- Installation and maintenance manual of the switches at delivery

10. GUARANTEE

The supplier should guarantees the R.M.U against all defects arising out of faulty designed or workmanship or of defective materials for a period of 12 months from putting in service, or 18 months after delivery, whichever expires earlier. The good quality and manufacturing of the materials should be guaranteed. Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge

11. GUARANTEE TABLES

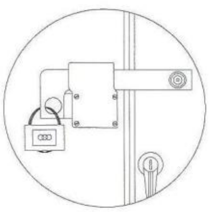
- The tenderer should fill in the attached guarantee tables.
- Any tender not accompanied with clear and complete guarantee tables should be rejected.

١٢. في حالة اللوحة خارج المبنى يلزم إضافة الاتي:

- توضع اللوحة داخل صندوق معدني لحمايتها من العوامل الجوية وذلك من صاج بسمك ٢.٥ مم قبل الدهان ويدهن ببوية بودرة الكتروستاتيك لون رمادي فاتح (يمكن جلفنه اللوحه بالكامل قبل الدهان ببودره الالكتروستاتك كبند اختياري حسب متطلبات شركه التوزيع) وتقفل اللوحة الخارجية بباب ضلفتين (بحيث ترتفع بداية الضلف عن قاعدة اللوحة بمسافة لا تقل عن ٢٠ سم)على الخلايا جميعها بواسطة قفل مسوجر MASTER KEY بعدد٣ مفاتيح ، كما يركب ترباس حديد كبير مجلفن علوي وآخر سفلي بالدلفة اليسرى بحيث لا يمكن فتح الدلفة إلا بمعرفة مختص من الشركة للعمل باللوحة ويكون للوحة الخارجية سقف منحدر علي ٦ درجات مائله لتصريف مياه الأمطار ، كما يخصص بالباب

الخارجي للوحة مكان لتركيب قفل عليها (رزة) وتركيب جوانات الحماية الكافية لتحقيق درجة حماية IP 54 .

- تجهز اللوحة بقاعدة من الكمر الحديد مناسب لوزن اللوحة مع تجهيز الكمر لرفع اللوحة من اسفل ويدهن الكمر بطبقتين احدهما من البرايمر
- تجهز الأبواب الخارجية بعدد ٢ رزة خارجية كما بالرسم وعدد 1 رزة لتركب عليها اقفال



- باقي المواصفات وكذلك البعد بين الأجزاء التى عليها جهد وجسم اللوحة تكون طبقا للمواصفات القياسية"(IEC 62271-200)– وطبقا لاصول الصناعة. وتكون الأبعاد بين الفاز والفاز الآخر وبين الفاز والأرضي طبقا للمواصفات الفنية المرفقة سابقا وايضا طبقا "(IEC 60071)".
- درجة الحماية IP54 ويجب أن يزود كل باب من الأبواب الداخلية والخارجية بـ RUBBER SEALING GASKET
- وسيلة عدم غلق الأبواب الخارجية أثناء الصيانة تكون بواسطة قضيب من الصلب مثبت فى جسم اللوحة بمسمار من أحد طرفيه وبه ثقب من الطرف الآخر يتم تثبيته ببنز مثبت بضلفة الباب الخارجي عند فتحه.
- يلزم ان يكون سقف اللوحة الخارجيه مجلفن علي الساخن ثم يتم دهانه الكتروستاتيك.

١٣. ملاحظات عامة :

- سكينة الفصل على الحمل(للكابل) من النوع الذى يفصل دائرياً أو النوع الذى يجمع ما بين الرأسي والدائري وسيعتبر هذا النوع الأخير كأنه دائري وسعة السكينة ٦٣٠ أمبير وجهد التشغيل 22 ك. ف. وسعة القطع لا تقل عن 20 ك.أ / ثانية لكل نوع من النوعين السابقين وتزود كل سكينة من السكاكين: بثلاثة مسامير مجلفنه كاملة بالصواميل والورد والورد السوستة لربط أطراف نهاية الكابل بها كما تزود كل سكينة من السكاكين الثلاثة بسكينة أرضي تعمل يدوياً مع وجود رباط ميكانيكي بين السكينة الأساسية وسكينة الأرضي يمنع توصيل السكينة الأساسية في حالة توصيل سكينة الأرضي وكذلك يمنع توصيل سكينة الأرضي في حالة توصيل السكينة الأساسية وتكون السكاكين بالمواصفات المرفقة سابقا.
- بالنسبة لسكينة الفصل على الحمل (للمحول) من النوع الذي يفصل دائرياً أو النوع الذي يجمع

ما بين الرأسي والدائري وسيعتبر هذا النوع الأخير كأنه دائري وسعة السكينة ٤٠٠ أمبير وجهد التشغيل 22 ك.ف. وسعة القطع لا تقل عن 20 ك.أ/ ثانية لكل نوع من النوعين السابقين، كاملة بثلاثة قواعد مصهرات 24 ك.ف. معزولة في حالة ان المسافة بين الفازات اقل من 22 سم مجهزة ميكانيكياً بالأذرع التي تفصل السكينة في حالة احتراق أحد المصهرات وتزود السكينة بثلاثة مسامير مجلفنة كاملة بالصواميل والورد والورد السوستة لربط أطراف نهاية كابل المحول بها وتكون السكاكين بالمواصفات المرفقة سابقا.

- يلزم توريد عدد (١) واحد يد تشغيل للسكاكين بذراع بطول مناسب للتشغيل لكل لوحة.
- يلزم توريد جهاز بيان التسرب الارضي كامل بمحول تيار من النوع الذي يركب على الغلاف الخارجي للكابل. كما يركب كل جهاز داخل علبة من الصاج في الجزء العلوي من اللوحة أو على الجوانب والعلبة تكون بفتحة لرؤية راية الجهاز عند سقوطها ويكون الجهاز Automatic reset حسب المواصفات المرفقة سابقا.
- جميع التوصيلات الداخلية الواصلة من محولات التيار إلى أجهزة بيان التسرب الأرضي تكون داخل مواسير بلاستيك P.V.C مثبتة بصاج اللوحة من الداخل .
- تكون اللوحة كاملة ببار الأرضي ويتم توصيل كل سكينة من سكاكين الأرضي لهذا البار بواسطة شيلد نحاس مقطعة ٢٥ مم٢.
- كل سكينة تكون فى خلية مستقلة داخل اللوحة كما يكون لكل سكينة دلفة خاصة بها مزودة بمفصلات جيدة مجلفنة على أن تفتح أفقيا بالنسبة لسكاكين الكابلات وتقفل الدلفة بواسطة عدد (٢) قفل مسوجر MASTER KEY
- تزود كل خلية بقفيز به مسمارين كاملين بالصواميل والورد المجلفنة لتثبيت الكابل أو صندوق نهاية الكابل بها وذلك أسفل السكينة وأعلى قاعدة الخلية أو أى وسيلة تؤدى الغرض.
- وتكون المسافات بين مسمار تثبيت الكابل بالسكينة وفتحة دخول الكابل للوحة مناسبة لتركيب نهايات الكابلات 18/30 وطبقا لأصول الصناعة .
- يقدم مع العطاء رسم تنفيذي مفصل بجميع البيانات والمواصفات التى يعرضها المورد وأيضا كتالوجات لكل المهمات المركبة بالخلية.
- يقدم المورد شهادة بالاختبارات النوعية الخاصة بالمهمات الموردة طبقا للمواصفات القياسية " IEC "

- يتم عمل الاختبارات الروتينية فى حضور مندوب الشركة طبقا للمواصفات القياسية " IEC "
- يتم توصيل محولات التيار ومحولات الجهد بروزته لتكون جاهزة لتركيب عدادات الطاقة الفعالة والغير فعالة (K.WH) + (KVARH) باللوحة (في حاله ان اللوحة مجهزة بمحولات تيار وجهد).
- يلزم ان تكون محولات التيار والجهد (في حاله ان اللوحة مجهزة بمحولات تيار وجهد) من الانواع المعتمدة لدي الشركة القابضة لكهرباء مصر.
- في حاله ان دخول الكابل لسكينه المحول من الجانب :يلزم تركيب علبه من الصاج لتغطيه وحماية الكابل.
- قطع الغيار(بند اختياري) : يلزم تقديم عرض سعر منفصل لقطع الغيار اللازمه للسكاكين (الميكانيزم+ اذرع نقل الحركة +اسلحة السكينه).
- يلزم عمل إنترلوك ميكانيكى يمنع فتح الباب إلا بعد توصيل سكينة الأرضى (بند اختياري علي حسب متطلبات شركات التوزيع)
- مسار التسريب الارضي للعازلات ومحولات التيار والجهد وذراع السكينه المركبه ٢سم/ك ف
- يتم دهان اللوحة خارج المبني stone paint بالنسبة للوحات المستخدمة بالمناطق الساحلية (بند اختياري علي حسب متطلبات شركات التوزيع)
- الالتزام بان يكون سمك الدهانات للوحات سواء داخل المبني او خارج المبني طبقا للـ 2- iso 12944 علي ان لا تقل درجة التأكل للوحات C4 وسمك الدهانات يتراوح (60 µm : 80 µm) او حسب متطلبات شركات التوزيع

Corrosivity category	Mass loss per unit surface/thickness loss (after first year of exposure)				Examples of typical environments in a temperate climate (informative only)	
	Low-carbon steel		Zinc		Exterior	Interior
	Mass loss g/m²	Thickness loss µm	Mass loss g/m²	Thickness loss µm		
C1 very low	⩽ 10	⩽ 1,3	⩽ 0,7	⩽ 0,1	—	Heated buildings with clean atmospheres, e.g. offices, shops, schools, hotels.
C2 low	> 10 to 200	> 1,3 to 25	> 0,7 to 5	> 0,1 to 0,7	Atmospheres with low level of pollution. Mostly rural areas.	Unheated buildings where condensation may occur, e.g. depots, sports halls.
C3 medium	> 200 to 400	> 25 to 50	> 5 to 15	> 0,7 to 2,1	Urban and industrial atmospheres, moderate sulfur dioxide pollution. Coastal areas with low salinity.	Production rooms with high humidity and some air pollution, e.g. food-processing plants, laundries, breweries, dairies.
C4 high	> 400 to 650	> 50 to 80	> 15 to 30	> 2,1 to 4,2	Industrial areas and coastal areas with moderate salinity.	Chemical plants, swimming pools, coastal ship- and boatyards.
C5-I very high (industrial)	> 650 to 1 500	> 80 to 200	> 30 to 60	> 4,2 to 8,4	Industrial areas with high humidity and aggressive atmosphere.	Buildings or areas with almost permanent condensation and with high pollution.
C5-M very high (marine)	> 650 to 1 500	> 80 to 200	> 30 to 60	> 4,2 to 8,4	Coastal and offshore areas with high salinity.	Buildings or areas with almost permanent condensation and with high pollution.

NOTES

1 The loss values used for the corrosivity categories are identical to those given in ISO 9223.

2 In coastal areas in hot, humid zones, the mass or thickness losses can exceed the limits of category C5-M. Special precautions must therefore be taken when selecting protective paint systems for structures in such areas.

Guarantee Table (1) For CABLE L.B.S

No	Item	Specif.
1	Type and manufacturer	
2	Nominal operating voltage	
3	Maximum operating voltage	
4	Nominal current at40°c	
5	Thermal withstand current for one sec (r.m.s)	
6	Making current (peak)	
7	Minimum clearances in air:	
	• between phases	
	• between phase and earth	
	• pole center	
8	Transformer off-load breaking capacity (PF=0.15)	
9	Capacitive breaking current	
10	▪ Technical specifications for the Earthing switch: - Making current (peak) -Mechanical endurance class -Thermal withstand current for one sec(r.m.s)	

We guarantee the data given above for the equipment offered.

Signature: ……………………………… **Date: ………………………………**

Guarantee Table (2) For TRANSFORMER L.B.S

No	Item	Specif.
1	Type and manufacturer	
2	Nominal operating voltage	
3	Maximum operating voltage	
4	Nominal current at 40°c	
5	Thermal withstand current for one sec (r.m.s)	
6	Making current (peak)	
7	Minimum clearances in air between phases at least	
8	Minimum clearances in air Pole center at least	

We guarantee the data given above for the equipment offered.

Signature: ………………………………

Date: ………………………………

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Guarantee Table (3) For ENCLOSURE & FUSE

Item	Enclosure
- Maker's name	
- Type	
- Material	
- Thickness of steel sheet (Inside and outside)	
- Degree of protection	
-Thickness of coatings	
- Dimensions: Width* Depth* Height	
Item	**Fuse**
Type	
- Rated voltage	
- Rated frequency	
- Rated current	
- Rated breaking capacity	

We guarantee the data given above for the equipment offered.

Signature: ………………………………

Date: ……………………………

Guarantee Table (4) For BUS BAR & insulation level

Item	Bus bars
Material	
Nominal operating voltage	
Maximum operating voltage	
Nominal current for Main bus bars & connectors	
Thermal withstand current for one sec (r.m.s.)	
Dynamic withstand current (peak)	
C.S.A of the main bus bar & connectors	
C.S.A of the earthing bus bar	
Material purity	
Material Conductivity	
Item	**INSULATION LEVEL**
One minute, power frequency withstand voltage, r.m.s : • To earth and between phases. • Across isolating distance (for load-break switches and fuses).	
1.2/50 µs impulse withstand voltage, peak :	

• To earth and between phases. • Across isolating distance (for load-break switches and fuses).	
Leakage path of supporting insulators	
Heat Shrinkable tube insulation (Anti-track) 18/30 KV	YES / NO

We guarantee the data given above for the equipment offered.

Signature: ………………………………

Date: …………………………….….

Guarantee Table (5) for V.T, E.F.I & heaters

Item	Specif.
1. For potential transformer :	
Type and maker name .	
No.	
Rated voltage .	
Burden and accuracy .	
Leakage path	
2. E.F.I	
No.	
Rated operating current	
Rated operating voltage	
3. Heaters	
Rated power	Watt
Hygrostat	Yes/No
M.C.B	Yes/No

We guarantee the data given above for the equipment offered.

Signature: ……………………………..

Date: ………………………………….

Guarantee Table (6) for general data for LBS

Item	Specif.
▪ The contact resistance per phase	
▪ Opening time for load break switch	
▪ Minimum Mechanical endurance Operating cycle M1	YES/NO
▪ Minimum Electrical endurance class E3	YES/NO
▪ Capacitive current breaking switch class C2	YES/NO
▪ The thickness of galvanizing coating for all steel parts should be not less than 12 μm.	YES/NO
▪ The terminals of connection should be silver-plated with thickness not less than 5μm.	YES/NO

We guarantee the data given above for the equipment offered.

Signature: ………………………………

Date: ………………………………….